

A Gaussian Surrogate of Partially Observed Stochastic Processes using Wasserstein Metric

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Abstract

Approximating the evolution of probability measures for nonlinear stochastic differential equations (SDEs) and the associated nonlinear filtering problems is a challenging problem as it involves solving high-dimensional differential equations. In contrast to classical variational inference methods which address this challenge by minimizing the Kullback-Leibler (KL) divergence between the true and approximate distributions, we propose a Wasserstein-based variational framework for approximating the laws of stochastic systems. In particular, instead of minimizing the KL divergence, our approach minimizes the Wasserstein-2 (W_2) distance between the joint probability distributions of the state and observation processes. This formulation respects the underlying transport geometry and results in evolution equations for Gaussian parameters that provide an approximation of the dynamics of the true measure. An illustration is provided for some of our results with the help of an academic example.

I. INTRODUCTION

For nonlinear stochastic dynamical systems, describing the evolution of the underlying probability measure is a challenging task. The probability law of the system evolves according to the Kolmogorov forward equation (also known as the Fokker-Planck equation) [1], [2], a partial differential equation (PDE) whose analytical solution is available only in very special cases. In practice, the evolution of the probability measure is intractable for most nonlinear and non-Gaussian systems. A closely related problem arises in nonlinear filtering, where one seeks to infer the posterior distribution of the system state given noisy observations. The evolution of this conditional law is governed by stochastic partial differential equations (SPDEs) such as the Kushner-Stratonovich and Zakai equations [3]–[5]. Solving these SPDEs exactly is rarely feasible, which motivates the search for approximation techniques that can capture the essential behavior of the evolving probability measures.

Several methods have been proposed for computing approximate solutions to the evolution of probability measures associated with stochastic processes. Among these, Gaussian approximations—where the true measure is projected onto the family of Gaussian distributions—have proved particularly popular [6], [7]. Such approaches lead to moment closure schemes or variational formulations where the mean and covariance evolve according to ordinary differential equations derived from minimizing an appropriate divergence measure. In this context, variational inference (VI) provides a principled framework for approximating intractable probability distributions by tractable surrogates. In the classical setting, the approximation is obtained by minimizing the Kullback–Leibler (KL) divergence between the true distribution and the approximating family. The KL-based variational formulation has been applied to both stochastic differential equations (SDEs) and the nonlinear filtering problem [8]–[10], yielding efficient Gaussian and mixture-Gaussian approximations to the corresponding posterior or marginal distributions. These approaches have also found wide application in Bayesian inference, control, and signal processing [11].

However, the KL divergence has inherent limitations when used for approximating dynamical systems. It is asymmetric and primarily sensitive to differences in low-probability regions, which can lead to biased approximations—particularly in the tails or in multimodal settings [12]. Moreover, minimizing the KL divergence often corresponds to moment matching or energy functional minimization that does not directly reflect the geometry of the underlying probability space. As an alternative to finding approximations using KL divergence, we take a different perspective in this paper and consider a variational approach based on Wasserstein-2 (W_2) distance. That is, instead of the KL divergence, we measure the discrepancy between probability measures using

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the W_2 distance. The W_2 distance arises naturally while studying optimal transport problems where the transportation cost between two prespecified marginals is described by a quadratic function. In contrast to other distance measures between probability distributions, such as the total variation distance, the W_2 distance defines a geometrically meaningful metric on the space of probability measures. This property has made it particularly valuable for a wide range of data science applications (see, for example, [13]). We refer to [14]–[16] for textbook treatments on optimal transport.

Our contribution differs from classical VI-based approaches in two aspects. First, we propose to approximate the joint probability distribution of the state and observation processes, instead of concentrating exclusively on the conditional posterior, as in [17]. This formulation provides a more global view of the system dynamics and avoids separately handling normalization or marginalization inherent in conditional distributions. Second, we obtain closed form relations for linear surrogate by minimizing the W_2 distance between the joint distributions, leading to a Gaussian approximation that is consistent with the natural geometry of the space of probability measures. This is enabled by our continuous–discrete setup: an Euler–Maruyama step makes the conditional laws Gaussian even under nonlinear dynamics. Since W_2 distance between Gaussians has an explicit formula, we can solve the variational problem analytically, without solving a separate optimal-transport problem at each iteration. While the use of W_2 metric between joint distributions is not seen in the literature, we see that the recent work [18] provides Wasserstein-1 bounds on the gap between the true joint distribution and its moment-matched Gaussian approximation in nonlinear filters.

The remainder of the paper is organized as follows. Section II formulates the problem of designing a Gaussian surrogate using Wasserstein-based variational formulation and introduces some relevant notions from optimal transport. In Section III, we solve the problem for the fully observed SDEs case and present an illustrative example. Section IV addresses the partially observed SDEs and derives an approximation of the posterior distribution. Finally, Section V concludes the paper and discusses future extensions.

II. PROBLEM SETTING

Let us consider a probability space $(\Omega, \mathcal{F}, \mathbb{P})$ and the standard Wiener process $(\beta_t)_{t \geq 0} \in \mathbb{R}^m$ with infinitesimal covariance equal to identity, that is, $\mathbb{E}[d\beta_t d\beta_t^\top] = I_m dt$. The state process of our interest is a continuous-time Markov process $(X_t)_{t \geq 0}$ in $\mathbb{R}^n$ whose evolution is described by the following stochastic differential equation (SDE):

$$dX_t = f(t, X_t) dt + G(t) d\beta_t, \quad X_0 \sim \rho_0 \quad (1)$$

where $f : \mathbb{R} \times \mathbb{R}^n \rightarrow \mathbb{R}^n$ describes the drift term, and the diffusion term $G : \mathbb{R} \rightarrow \mathbb{R}^{n \times m}$ is assumed to be time-dependent only, and X_0 is independent of $(\beta_t)_{t \geq 0}$. Under standard regularity assumptions (e.g., global Lipschitz and linear growth in x and $G(\cdot)$ being locally bounded), there exists unique strong solution $(X_t)_{t \geq 0}$ [2, Theorem 5.2.1]. Note, due to the nonlinearity in f , the process need not be Gaussian even when ρ_0 is Gaussian.

As noted earlier, we consider the approximation with respect to the W_2 distance between distributions. In what follows, we first overview some fundamental notions from the theory of optimal transport to define W_2 distance. Then, we formulate the first optimization problem that provides an approximation of the distribution of the state process. We then associate an observation process with (1) and formulate an optimization problem to approximate the posterior of the state conditioned upon this observation process.

A. Optimal Transport Preliminaries

We briefly recall notation from optimal transport (see, e.g., [13], [14]).

Spaces and measures. Let $(\mathcal{X}, \|\cdot\|)$ be a Euclidean space (or a Polish metric space) and let $\mathcal{P}_2(\mathcal{X})$ be the set of Borel probability measures with finite second moment:

$$\mathcal{P}_2(\mathcal{X}) := \left\{ \mu : \int_{\mathcal{X}} \|x\|^2 d\mu(x) < \infty \right\}.$$

Pushforward. For a measurable map $T : \mathcal{X} \rightarrow \mathcal{Y}$ and $\mu \in \mathcal{P}_2(\mathcal{X})$, the *pushforward* measure $T_{\#}\mu \in \mathcal{P}_2(\mathcal{Y})$ is

$$T_{\#}\mu(O) := \mu(T^{-1}(O)), \quad O \subset \mathcal{Y} \text{ Borel}.$$

Couplings and marginals. For $\mu \in \mathcal{P}_2(\mathcal{X})$ and $\nu \in \mathcal{P}_2(\mathcal{Y})$, a *coupling* (or transport plan) is a probability $\gamma \in \mathcal{P}_2(\mathcal{X} \times \mathcal{Y})$ whose marginals are μ and ν :

$$\Gamma(\mu, \nu) := \{ \gamma : \pi_{1\#} \gamma = \mu, \pi_{2\#} \gamma = \nu \},$$

where $\pi_1(x, y) = x$ and $\pi_2(x, y) = y$ are the canonical projections, and $\pi_{i\#} \gamma$ denotes the pushforward (i.e., the i -th marginal).

Quadratic Wasserstein distance. For $\mu, \nu \in \mathcal{P}_2(\mathcal{X})$, the Wasserstein-2 (W_2) distance is defined as

$$W_2^2(\mu, \nu) := \inf_{\gamma \in \Gamma(\mu, \nu)} \int_{\mathcal{X} \times \mathcal{X}} \|x - y\|^2 d\gamma(x, y). \quad (2)$$

This defines a metric on $\mathcal{P}_2(\mathcal{X})$.

Product spaces and notation. For random vectors (U, V) on $\mathbb{R}^{d_1} \times \mathbb{R}^{d_2}$ with law P_{UV} , we write

$$P_U := \pi_{1\#} P_{UV}, \quad P_V := \pi_{2\#} P_{UV}.$$

Gaussian case (closed form and map). Let $\mu = \mathcal{N}(m_1, \Sigma_1)$ and $\nu = \mathcal{N}(m_2, \Sigma_2)$ be two non-degenerate Gaussian measures on $\mathbb{R}^n$. Then the squared W_2 distance between μ and ν admits the following closed form:

$$W_2^2(\mu, \nu) = \|m_1 - m_2\|^2 + \mathcal{B}^2(\Sigma_1, \Sigma_2), \quad (3)$$

where

$$\mathcal{B}^2(\Sigma_1, \Sigma_2) = \text{Tr} \left(\Sigma_1 + \Sigma_2 - 2(\Sigma_2^{1/2} \Sigma_1 \Sigma_2^{1/2})^{1/2} \right)$$

is the *squared Bures metric* between covariance matrices (see [19] for the closed form and [20] for modern treatment of Wasserstein-Bures metric). Moreover, when μ is absolutely continuous w.r.t. the Lebesgue measure (e.g., $\Sigma_1 \succ 0$), the unique W_2 -optimal transport map $\mathcal{T} : \mathbb{R}^n \rightarrow \mathbb{R}^n$ pushing μ to ν is affine:

$$\mathcal{T}(x) = m_2 + M(x - m_1), \text{ where } M := \Sigma_2^{1/2} (\Sigma_2^{1/2} \Sigma_1 \Sigma_2^{1/2})^{-1/2} \Sigma_2^{1/2} \quad (4)$$

(see [21] for the explicit map).

B. Gaussian Surrogate for Nonlinear SDE

To address the problem of computing a Gaussian approximation of process defined in (1) over an interval $[0, T]$, let us consider the Euler–Maruyama discretization scheme with step size τ . We choose a time instant $s \geq 0$, such that, for some $k \in \mathbb{N}$, we have $s = k\tau$. We use the notation x_s , or x_k when the time step τ is obvious from the context, to denote $x_{k\tau}$. The discretization, therefore, leads to the transition density

$$\rho(x_{s+\tau} | x_s) = \mathcal{N}(x_s + \tau f(x_s), \Sigma_x(s)), \quad (5)$$

where $\Sigma_x(s) := G(s)G(s)^\top \tau$.

To describe the family of approximating distributions σ , we consider a surrogate linear SDE that admits a Gaussian evolution, namely

$$dZ_t = (A(t)Z_t + b(t))dt + B(t)d\beta_t \quad (6)$$

with deterministic time-varying design parameters $A : \mathbb{R}_{\geq 0} \rightarrow \mathbb{R}^{n \times n}$, $B : \mathbb{R}_{\geq 0} \rightarrow \mathbb{R}^{n \times m}$, and the vector $b : \mathbb{R}_{\geq 0} \rightarrow \mathbb{R}^n$. Discretizing the linear system (6) with time step τ yields the Gaussian transition kernel

$$\sigma(z_{s+\tau} | z_s) = \mathcal{N}(z_s + \tau A(s)z_s + b(s)\tau, \Sigma_z(s)), \quad (7)$$

such that $\Sigma_z(s) := B(s)B(s)^\top \tau$.

The problem of finding an optimal linear surrogate for a SDE over a certain time interval can be formulated as finding optimal system parameters for each time step by minimizing the Wasserstein-2 distance defined in (2) between the nonlinear and linear joint laws at each time instant. In other words, let $\rho_x(x_{s+\tau}, x_s)$ denote the joint probability distribution between $x_{s+\tau}$ and x_s induced by the transition kernel in (5). The objective is to design the parameters in (6) so that the joint distribution $\sigma_z(z_{s+\tau}, z_s)$, induced by the transition kernel in (7), approximates $\rho_x(x_{s+\tau}, x_s)$ with

respect to the W_2 metric. In particular, for every time $s = k\tau$, $k \in \mathbb{N}$, we seek the unknown system data $A(s)$, $b(s)$ and $B(s)$ that minimize

$$\min_{A(s), b(s), B(s)} W_2^2(\rho_x(x_{s+\tau}, x_s), \sigma_z(z_{s+\tau}, z_s))$$

as the discretization step $\tau \rightarrow 0$. The solution to this problem is provided in Section III, where we construct a Gaussian process (6) in which the parameters $A(s)$, $b(s)$ and $B(s)$ are chosen to be optimal at each time step.

C. Gaussian Surrogate for Posterior

We next consider the problem of extending our approach for computing the Gaussian approximation of posterior distribution of the state conditioned upon an observation process. More precisely, with system (1), we associate an observation process

$$Y_{t_k} = h(X_{t_k}) + \epsilon_k, \quad (8)$$

where, for the sake of simplicity in this paper, we assume that $t_k = k\tau$, with $k \in \mathbb{N}$. The mapping $h : \mathbb{R}^n \rightarrow \mathbb{R}^p$ is assumed to be continuous, and $\epsilon_k \sim \mathcal{N}(0, R_k)$ is the mean-zero Gaussian noise with covariance $R_k \in \mathbb{R}^{p \times p}$ symmetric and positive definite, for each $k \in \mathbb{N}$.

The posterior distribution, which we want to approximate, is ideally determined using the Bayes rule:

$$\rho(x_{t_k} | y_{t_k}) \propto \rho(y_{t_k} | x_{t_k}) \rho(x_{t_k}), \quad (9)$$

where $\rho(y_{t_k} | x_{t_k})$ is the likelihood function, and $\rho(x_{t_k})$ is the prior distribution at time t_k . We consider the joint distribution over states at consecutive sampling instants, and the next observation, that is,

$$\rho_{xy}(x_{s+\tau}, x_s, y_{s+\tau}) \in \mathcal{P}(\mathbb{R}^{2n+m}).$$

By the disintegration theorem [22, Corollary 10.4.13], this joint law can be written as

$$\rho_{xy}(x_{s+\tau}, x_s, y_{s+\tau}) = \rho(y_{s+\tau} | x_{s+\tau}, x_s) \rho_x(x_{s+\tau}, x_s).$$

Under the Markovian assumption on the state process, the dependence simplifies to

$$\rho_{xy}(x_{s+\tau}, x_s, y_{s+\tau}) = \rho(y_{s+\tau} | x_{s+\tau}) \rho_x(x_{s+\tau}, x_s),$$

which can be further decomposed as

$$\rho_{xy}(x_{s+\tau}, x_s, y_{s+\tau}) = \rho(y_{s+\tau} | x_{s+\tau}) \rho(x_{s+\tau} | x_s) \rho(x_s). \quad (10)$$

To obtain Gaussian approximation for this posterior, we append the linear surrogate (6) with a linear observation process:

$$w_k = H_k Z_k + d_k + \delta_k, \quad (11)$$

so that the measurement w_k is available at time $s = k\tau$. Here, for each $k \in \mathbb{N}$, $\delta_k \sim \mathcal{N}(0, \tilde{R}_k)$ and we want to choose the positive definite covariance matrix $\tilde{R}_k$, the matrix $H_k \in \mathbb{R}^{p \times n}$, and the vector $d_k \in \mathbb{R}^n$ to compute the best possible Gaussian approximation of the posterior using this model. We next consider the joint distribution coming from the surrogate model (6) and (11)

$$\sigma_{zw}(z_{s+\tau}, z_s, w_{k+1}) = \sigma(w_{k+1} | z_{s+\tau}) \sigma(z_{s+\tau} | z_s) \sigma(z_s). \quad (12)$$

Define $\tilde{x} := (x_{s+\tau}, x_s)$ and $\tilde{z} := (z_{s+\tau}, z_s)$. The W_2 distance between the joint distributions σ_{zw} and ρ_{xy} is then given by

$$W_2^2(\rho_{xy}, \sigma_{zw}) = \inf_{\gamma \in \mathcal{P}(\mathbb{R}^{2n+m} \times \mathbb{R}^{2n+m})} \int (\|\tilde{x} - \tilde{z}\|^2 + \|y_{k+1} - w_{k+1}\|^2) d\gamma(\tilde{x}, y_{k+1}; \tilde{z}, w_{k+1}) \quad (13)$$

such that $\pi_{1\#}\gamma = \rho_{xy}$, $\pi_{2\#}\gamma = \sigma_{zw}$.

We propose projecting ρ_{xy} onto the Gaussian manifold using the Wasserstein metric, that is,

$$\min_{\sigma_{zw}} W_2(\sigma_{zw}, \rho_{xy}), \quad (14)$$

such that σ_{zw} is a Gaussian distribution. The minimization above is carried out w.r.t. the system parameters $A(s)$, $b(s)$, $B(s)$, H_k , d_k , $\tilde{R}_k$.

With the disintegration shown in (12), the problem in (14) naturally decomposes into a two-step prediction–update procedure. In particular, the projection decomposes into two parts: the first corresponds to the propagation step described earlier in Section III, while the second corresponds to the posterior projection step.

III. GAUSSIAN SURROGATE FOR STATE PROCESS

In this section, we address the problem of calculating the Gaussian surrogate for the nonlinear SDE (1), as defined in Section II-B, without taking the observation process into consideration.

We begin by recalling the notation. Let $(X_t)_{t \in [0, T]}$ solves the nonlinear SDE (1). For a fixed $\tau \geq 0$ such that $s = k\tau$ for $k \in \mathbb{N}$, the joint distribution between $X_{s+\tau}$ and X_s is denoted by ρ_x . Similarly, let $(Z_t)_{t \in [0, T]}$ solve the linear surrogate SDE (6) with system parameters $(A(s), B(s), b(s))$, and let the joint distribution between $Z_{s+\tau}$ and Z_s be denoted by σ_z . The W_2 distance between ρ_x and σ_z is given by

$$W_2^2(\rho_x, \sigma_z) = \inf_{\kappa \in \mathcal{P}(\mathbb{R}^{2n} \times \mathbb{R}^{2n})} \int \| (x_{s+\tau}, x_s) - (z_{s+\tau}, z_s) \|^2 d\kappa \quad (15)$$

such that $\pi_{1\sharp}\kappa = \rho_x$, $\pi_{2\sharp}\kappa = \sigma_z$,

where $\pi_i : \mathbb{R}^{2n} \rightarrow \mathbb{R}^n$, $i = 1, 2$, are the canonical projection mappings such that $\pi_1(z, x) = z$, and $\pi_2(z, x) = x$. We are now ready to solve the surrogate design problem which is expressed as

$$\min_{A(s), b(s), B(s)} W_2^2(\rho_x, \sigma_z). \quad (16)$$

In order to obtain computable approximations for ρ , at each time step we approximate the true prior distribution $\rho \in \mathcal{P}(\mathbb{R}^n)$ with a Gaussian distribution $\sigma \in \mathcal{P}(\mathbb{R}^n)$. This approximation is characterized by a mean $\mu_s = \mathbb{E}_\sigma(Z_s)$ and a covariance $\Sigma_s = \mathbb{E}_\sigma(Z_s Z_s^\top)$, so that $\sigma = \mathcal{N}(\mu_s, \Sigma_s)$.

A. Main result

We can now state the main result.

Theorem 1. *The minimizer of (16) is unique and given by*

$$A^*(s) = C(s) \Sigma_s^{-1}, \quad (17)$$

$$b^*(s) = \phi(s) - A^*(s) \mu_s, \quad (18)$$

$$B^*(s) B^*(s)^\top = G(s) G(s)^\top, \quad (19)$$

where $\phi(s) := \mathbb{E}_\sigma[f(s, X_s)]$, and $C(s) := \mathbb{E}_\sigma[(f(s, X_s) - \phi_s)(X_s - \mu_s)^\top]$.

In order to prove our main result, we need the following technical lemma.

Lemma 2. *The W_2 distance between $\sigma_z \in \mathcal{P}(\mathbb{R}^{2n})$ and $\rho_x \in \mathcal{P}(\mathbb{R}^{2n})$ in (16) is given by*

$$W_2^2(\rho_x(x_{s+\tau}, x_s), \sigma_z(z_{s+\tau}, z_s)) = \int_{\mathbb{R}^n} W_2^2(\rho(x_{s+\tau}|x_s), \sigma(z_{s+\tau}|x_s)) d\sigma(x_s). \quad (20)$$

Proof. Let $\Gamma(\rho_x, \sigma_z)$ denote the set of couplings between the two distributions ρ_x and σ_z . Take any $\kappa \in \Gamma(\rho_x(x_{s+\tau}, x_s), \sigma_z(z_{s+\tau}, z_s))$. We can disintegrate κ as

$$d\kappa(x_s, x_{s+\tau}; z_s, z_{s+\tau}) = d\zeta(x_s, z_s) d\xi(x_{s+\tau}, z_{s+\tau} | x_s, z_s),$$

where ζ is a coupling between the marginals $\rho(x_s)$ and $\sigma(z_s)$, and $\xi(\cdot, \cdot | x_s, z_s)$ is a coupling between the conditionals $\rho(x_{s+\tau}|x_s)$ and $\sigma(z_{s+\tau}|z_s)$. The quadratic transport cost between $(x_s, x_{s+\tau})$ and $(z_s, z_{s+\tau})$ is defined as

$$c((x_s, x_{s+\tau}), (z_s, z_{s+\tau})) := \|x_s - z_s\|^2 + \|x_{s+\tau} - z_{s+\tau}\|^2,$$

which gives

$$\mathbb{E}_\kappa[c] = \int d\zeta(x_s, z_s) \left[\|x_s - z_s\|^2 + \int \|x_{s+\tau} - z_{s+\tau}\|^2 d\xi(x_{s+\tau}, z_{s+\tau} | x_s, z_s) \right].$$

The second term in the expression above is bounded below by the optimal transport cost between $\rho(x_{s+\tau}|x_s)$ and $\sigma(z_{s+\tau}|z_s)$, since ξ is any admissible coupling between the two. Therefore, for every admissible couplings κ, ζ , and ξ , we have

$$\mathbb{E}_\kappa[c] \geq \iint \|x_s - z_s\|^2 d\zeta(x_s, z_s) + \int W_2^2(\rho(x_{s+\tau}|x_s), \sigma(z_{s+\tau}|z_s)) d\zeta(x_s, z_s). \quad (21)$$

The first term in (21) is nonnegative and vanishes under the identity coupling

$$d\zeta^*(x_s, z_s) = d\sigma(x_s) \delta_{x_s}(z_s),$$

since $\rho(x_s) = \sigma(z_s)$.

Fixing this choice of ζ^* , for each x_s we select $\xi_{x_s}^*$ as the optimal coupling between $\rho(x_{s+\tau}|x_s)$ and $\sigma(z_{s+\tau}|z_s)$. Then the product measure

$$d\kappa^*(x_s, x_{s+\tau}; z_s, z_{s+\tau}) = d\sigma(x_s) \delta_{x_s}(dz_s) d\xi_{x_s}^*(x_{s+\tau}, z_{s+\tau}) \quad (22)$$

belongs to $\Gamma(\rho_x, \sigma_z)$ and attains the right-hand side of (20). By inequality (21), this cost is minimal, which shows that κ^* is indeed optimal. Therefore, equality in (20) follows. $\square$

Proof of Theorem 1. Recall that $\rho(x_{s+\tau}|x_s)$ and $\sigma(z_{s+\tau}|z_s)$ are Gaussian distributions with laws given by (5) and (7), respectively. Substituting the exact expression (3) for the squared W_2 distance between the two Gaussian distributions into the RHS of (20) transforms the minimization problem (16) into

$$\min_{A(s), b(s), B(s)} W_2^2(\rho_x, \sigma_z) = \min_{A(s), b(s), B(s)} \tau^2 \int \sigma(dx) \|f(s, x) - A(s)x - b(s)\|^2 + \tau \text{Tr} \left[\Sigma_p(s) + \Sigma_q(s) - 2(\Sigma_p(s)^{1/2} \Sigma_q(s) \Sigma_p(s)^{1/2})^{1/2} \right], \quad (23)$$

where

$$\Sigma_p := G(s) G(s)^\top \tau, \quad \Sigma_q := B(s) B(s)^\top \tau.$$

Let us denote the objective by

$$\mathcal{J}(A(s), b(s), B(s)) = W_2^2(\rho_x(x_{s+\tau}, x_s), \sigma_z(z_{s+\tau}, z_s)).$$

(1) *Optimization w.r.t. $b(s)$* : The drift term of $\mathcal{J}$ is given by

$$\mathcal{J}_d(A(s), b(s)) := \tau^2 \mathbb{E}_\sigma[\|f(s, X_s) - A(s)X_s - b(s)\|^2],$$

which is the only term of $\mathcal{J}$ that depends on $A(s)$ and $b(s)$. To find the optimality condition, we differentiate $\mathcal{J}_d$ w.r.t. $b(s)$:

$$\frac{\partial \mathcal{J}_d}{\partial b(s)} = -2\tau^2 \mathbb{E}_\sigma[f(s, X) - A(s)X_s - b(s)] = 0.$$

Solving for $b(s)$, we find the optimal value

$$b^*(s) = \mathbb{E}_\sigma[f(s, X_s)] - A(s) \mathbb{E}_\sigma[X_s]. \quad (24)$$

(2) *Optimization w.r.t. $A(s)$* : Differentiating $\mathcal{J}_d$ w.r.t. $A(s)$ gives

$$\frac{\partial \mathcal{J}_d}{\partial A(s)} = -2\tau^2 \mathbb{E}_\sigma[(f(s, X) - A(s)X_s - b(s))X_s^\top] = 0.$$

Substituting $b(s)$ from (24) and simplifying, we obtain

$$A(s) \Sigma_s = C(s) \implies A(s) = C(s) \Sigma_s^{-1}, \quad (25)$$

where

$$C(s) = \mathbb{E}_\sigma[(f(s, X_s) - \phi_s)(X_s - \mu_s)^\top], \quad \phi_s = \mathbb{E}_\sigma[f(s, X_s)].$$

If Σ is singular, the Moore–Penrose pseudo-inverse is used.

(3) *Optimization w.r.t. $B(s)$* : The diffusion contribution to the objective is given by the *Bures distance*

$$\mathcal{B}(\Sigma_p, \Sigma_q) = \text{Tr} \left[\Sigma_p + \Sigma_q - 2(\Sigma_p^{1/2} \Sigma_q \Sigma_p^{1/2})^{1/2} \right].$$

Consider a variation of $\mathcal{B}$ with a symmetric perturbation $\Delta\Sigma_q$. Then,

$$\Delta\mathcal{B} = \text{Tr}\left[\Delta\Sigma_q - \Sigma_p^{1/2}(\Sigma_p^{1/2}\Sigma_q\Sigma_p^{1/2})^{-1/2}\Sigma_p^{1/2}\Delta\Sigma_q\right].$$

Setting $\Delta\mathcal{B} = 0$ for all $\Delta\Sigma_q$ implies

$$I_n - \Sigma_p^{1/2}(\Sigma_p^{1/2}\Sigma_q\Sigma_p^{1/2})^{-1/2}\Sigma_p^{1/2} = 0, \implies \Sigma_q = \Sigma_p.$$

Therefore, we obtain

$$B^*(s)B^*(s)^\top = G(s)G(s)^\top. \quad (26)$$

Combining (24), (25), and (26) yields the unique minimizers given in (17)–(19). $\square$

B. Corollary and Illustration

The optimal system data $(A^*(s), b^*(s), B^*(s))$ can then be used to propagate the mean and covariance of the linear model as follows:

$$\begin{aligned} \mu_{s+\tau} &= \mu_s + \tau \mathbb{E}_\sigma[f(s, X_s)], \\ \Sigma_{s+\tau} &= \Sigma_s + \tau \mathbb{E}_\sigma[f(s, X_s)(X_s - \mu_s)^\top] + \tau \mathbb{E}_\sigma[(X_s - \mu_s)f(s, X_s)^\top] \\ &\quad + \tau^2 \mathbb{E}_\sigma[f(s, X_s)(X_s - \mu_s)^\top] \Sigma_s^{-1} \mathbb{E}_\sigma[(X_s - \mu_s)f(s, X_s)^\top] + \tau G(s)G(s)^\top. \end{aligned} \quad (27)$$

Taking the limit $\tau \rightarrow 0$, we find a set of two coupled ODEs:

$$\begin{aligned} \dot{\mu}(t) &= \mathbb{E}_\sigma[f(t, X)], \\ \dot{\Sigma}(t) &= \mathbb{E}_\sigma[f(t, X)(X - \mu)^\top] + \mathbb{E}_\sigma[(X - \mu)f(t, X)^\top] + G(t)G(t)^\top. \end{aligned} \quad (28)$$

In this section, we present an example to illustrate the performance of the variational approximation method introduced, by comparing a nonlinear SDE with its linear approximation.

Example 1. We consider a one-dimensional nonlinear SDE of the form

$$dX_t = \sin(X_t) dt + 0.3 d\beta_t.$$

Using the results obtained in Section III, specifically (17), (18), (19), we obtain a linear surrogate. To write the expressions explicitly, we make use of the fact that, for a Gaussian $\sigma \sim \mathcal{N}(\mu_t, \Sigma_t)$, we have $\mathbb{E}_\sigma(f(x_t)) = \mathbb{E}_\sigma(\sin(x_t)) = e^{-\frac{1}{2}\Sigma_t} \sin(\mu_t)$. Similarly, $\mathbb{E}_\sigma(\cos(x_t)) = e^{-\frac{1}{2}\Sigma_t} \cos(\mu_t)$. Moreover, as $f(x_t) = \sin(x_t)$ is a differentiable function, then by Stein's Lemma [23], we have $\mathbb{E}_\sigma(\sin(x_t)(x_t - \mu_t)) = \Sigma_t \mathbb{E}_\sigma(\cos(x_t))$. Using these expressions, we arrive at

$$\begin{aligned} A(t) &= \mathbb{E}_\sigma(\cos(x_t)), \\ b(t) &= \mathbb{E}_\sigma(\sin(x_t)) - A(t)\mu_t, \\ B(t) &= G(t) = 0.3. \end{aligned}$$

Thus, the linear approximating SDE becomes

$$dX_t = [\mathbb{E}_\sigma(\cos(X_t))X_t + \mathbb{E}_\sigma(\sin(X_t)) - \mathbb{E}_\sigma(\cos(x_t))\mu_t] dt + 0.3 d\beta_t.$$

Using (27) we find the following expressions for the propagation of mean and covariance:

$$\begin{aligned} \mu_{s+\tau} &= \mu_s + (\mathbb{E}_\sigma(\sin(x_s))) \tau, \\ \Sigma_{s+\tau} &= \Sigma_s + \tau [2\Sigma_s \mathbb{E}_\sigma(\cos(x_s)) + 0.9]. \end{aligned}$$

We then simulate both processes: the nonlinear SDE and its linear approximation over the same time interval using the Euler–Maruyama scheme with time step $\tau = 0.01$. When this simulation is repeated five times with independent Brownian noise realizations, the linear trajectories closely track the nonlinear ones, demonstrating that our linear surrogate provides an accurate local Gaussian approximation to the nonlinear dynamics during propagation. The gray shaded region ($\pm 2\sigma$ avg) represents the 95% confidence interval predicted by the Gaussian covariance Σ_t , showing the region where the true nonlinear trajectories are expected to lie with high probability. This is shown in Figure 1.

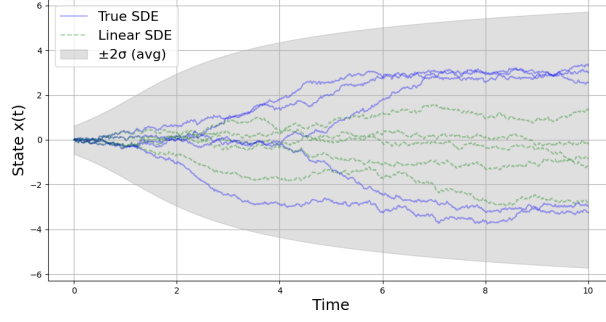


Fig. 1: Nonlinear SDE vs its Linear Approximation (5 Simulations).

IV. FROM PROPAGATION TO POSTERIOR

Next, we address the problem described in Section II-C of obtaining a Gaussian approximation of the posterior distribution in (9). We focus our attention on time-discretized version of the problem, where we fix a step size $\tau > 0$ and use the subscript $k \in N$ to denote the time instant $s = k\tau$; similarly, $k+1$ refers to the time instant $s + \tau$. To this end, we formulate a minimization problem over the system parameters $(A(s), b(s), G(s), H_k, d_k, \tilde{R}_k)$, with a squared W_2 objective measuring the distance between the joint distributions $\rho_{xy} \in \mathcal{P}(\mathbb{R}^{2n+p})$ and $\sigma_{zw} \in \mathcal{P}(\mathbb{R}^{2n+p})$. For notational convenience, let $\tilde{x} := (x_{k+1}, x_k)$ and $\tilde{z} := (z_{k+1}, z_k)$, and define $\rho_x(\tilde{x}) := \rho_x(x_{k+1}, x_k)$ and $\sigma_z(\tilde{z}) := \sigma_z(z_{k+1}, z_k)$. We disintegrate any admissible coupling γ as

$$d\gamma(\tilde{x}, y; \tilde{z}, w) = d\eta(y, w \mid \tilde{x}, \tilde{z}) d\kappa(\tilde{x}, \tilde{z}), \quad (29)$$

where $\kappa \in \Gamma(\rho_x, \sigma_z)$, and for each $(\tilde{x}, \tilde{z})$, $\eta(\cdot, \cdot \mid \tilde{x}, \tilde{z})$ is a coupling between $\rho(y_{k+1} \mid \tilde{x})$ and $\sigma(w_{k+1} \mid \tilde{z})$.

Substituting the disintegrated measure (29) into (13), we obtain

$$W_2^2(\rho_{xy}, \sigma_{zw}) = \inf_{\kappa} \int_{\mathbb{R}^{2n} \times \mathbb{R}^{2n}} \left\{ \|\tilde{x} - \tilde{z}\|^2 + \inf_{\eta} \int_{\mathbb{R}^p \times \mathbb{R}^p} \|y - w\|^2 d\eta(y, w \mid \tilde{x}, \tilde{z}) \right\} d\kappa(\tilde{x}, \tilde{z}) \quad (30)$$

such that $\kappa \in \Gamma(\rho_x, \sigma_z)$, $\eta \in \Gamma(\rho(y_{k+1} \mid \tilde{x}), \sigma(w_{k+1} \mid \tilde{z}))$.

The second term inside the curly braces in (30) represents the conditional transport cost between the distributions of the observation models, i.e., $W_2^2(\rho(y_{k+1} \mid \tilde{x}), \sigma(w_{k+1} \mid \tilde{z}))$. Thus, we can equivalently write the problem as

$$W_2^2(\rho_{xy}, \sigma_{zw}) = \inf_{\kappa} \int_{\mathbb{R}^{2n} \times \mathbb{R}^{2n}} \left\{ \|\tilde{x} - \tilde{z}\|^2 + \inf_{\kappa \in \Gamma(\rho_x, \sigma_z)} \int_{\mathbb{R}^p \times \mathbb{R}^p} W_2^2(\rho(y_{k+1} \mid \tilde{x}), \sigma(w_{k+1} \mid \tilde{z})) d\kappa(\tilde{x}, \tilde{z}) \right\} d\kappa(\tilde{x}, \tilde{z}). \quad (31)$$

In order to find a tractable iterative solution for the problem obtained after the substitution of (31) into (14), we adopt a two-stage approximation. First, we find κ^* which minimizes the first term on the RHS of (31), i.e., $\inf_{\kappa} \int_{\mathbb{R}^{2n} \times \mathbb{R}^{2n}} \|\tilde{x} - \tilde{z}\|^2 d\kappa(\tilde{x}, \tilde{z})$ which corresponds to $W_2^2(\rho_x, \sigma_z)$. We then use this κ^* as an approximation of the optimal coupling κ in (31), yielding

$$\min_{\Delta} W_2^2(\rho_{xy}, \sigma_{zw}) \leq \min_{\Delta} \left\{ W_2^2(\rho_x, \sigma_z) + \int_{\mathbb{R}^p \times \mathbb{R}^p} W_2^2(\rho(y_{k+1} \mid \tilde{x}), \sigma(w_{k+1} \mid \tilde{z})) d\kappa^*(\tilde{x}, \tilde{z}) \right\}, \quad (32)$$

where κ^* is the optimizer of the first term in (32), and $\Delta := (A(s), b(s), B(s), H_{k+1}, d_{k+1}, \tilde{R}_{k+1})$ denotes the collection of all design parameters.

The first term on the RHS of (32) depends only on $(A(s), b(s), B(s))$ and has already been calculated in Section III. This term can be seen as a propagation step, as it provides the optimal parameters for the linear system (6), which are then used to compute the updated mean and covariance as in (27).

In the next lemma, we derive the optimal coupling κ^* , which will be crucial in handling the second term on the RHS of (31).

Lemma 3. *The optimal coupling κ^* is given by*

$$d\kappa^*(x_{k+1}, x_k; z_{k+1}, z_k) = d\sigma_z(x_k) \delta_{x_k}(z_k) \times d\rho(x_{k+1}|x_k) \delta_{\mathcal{T}_{x_k}(x_{k+1})}(dz_{k+1}). \quad (33)$$

where $\mathcal{T}_{x_k}(x_{k+1}) = x_{k+1} - (A_k^* x_k + b_k^* - f(x))\tau$ and $x_{k+1} \sim \rho(\cdot|x_k)$.

Proof. We start with the disintegrated form of κ^* obtained in (22),

$$d\kappa^*(x_{k+1}, x_k; z_{k+1}, z_k) = d\sigma_z(x_k) \delta_{x_k}(dz_k) d\xi_{x_k}^*(x_{k+1}, z_{k+1}).$$

The optimal coupling $d\xi_{x_k}^*(x_{k+1}, z_{k+1})$ between two Gaussian conditionals

$$\begin{aligned} \rho(x_{k+1}|x_k) &= \mathcal{N}(x_k + f(x_k)\tau, \Sigma_p), \text{ and} \\ \sigma(x_{k+1} | x_k) &= \mathcal{N}(x_k + A_k^* x_k \tau + b_k^* \tau, \Sigma_q) \end{aligned}$$

will be

$$d\xi_{x_k}^*(x_{k+1}, z_{k+1}) = d\rho(x_{k+1}|x_k) \delta_{\mathcal{T}_{x_k}(x_{k+1})}(dz_{k+1}),$$

where the expression of $\mathcal{T}_{x_k}(x_{k+1})$ can be derived from (4). From Theorem 1, we have

$$B^*(s)B^*(s)^\top = G(s)G(s)^\top \Rightarrow \Sigma_p = \Sigma_q. \quad (34)$$

So in the Gaussian optimal transport map (4), we have $M = I$ and $\mathcal{T}_{x_k} = x_{k+1} + (A_k^* x_k + b_k^* - f(x_k))\tau$, where $b_k^* = \phi_k - A_k^* \mu_k$ as defined in 1. $\square$

A. Update step

Next, we focus on the second term on the RHS of (32). The Markovian nature of the dynamics in (8) and (11) implies that the coupling η , conditioned on $(\tilde{x}, \tilde{z}) = (x_{k+1}, x_k, z_{k+1}, z_k)$, depends only on x_{k+1} and z_{k+1} . Hence, we obtain

$$\begin{aligned} &W_2^2(\rho(y_{k+1}|\tilde{x}), \sigma(w_{k+1}|\tilde{z})) \\ &= \inf_{\eta} \int \|y_{k+1} - w_{k+1}\|^2 d\eta(y_{k+1}, w_{k+1} | x_{k+1}, z_{k+1}) \\ &= W_2^2(\rho(y_{k+1}|x_{k+1}), \sigma(w_{k+1}|z_{k+1})) \\ &\text{such that } \eta \in \Gamma(\rho(y_{s+\tau} | x_{s+\tau}), \sigma(w_{s+\tau} | z_{s+\tau})). \end{aligned}$$

Thus, the problem of finding a linear surrogate for the partially observed system, as presented in (14), can be written as

$$\min_{\Delta} W_2^2(\rho_{xy}, \sigma_{zw}) \leq \min_{\Delta} \left\{ W_2^2(\rho_x, \sigma_z) + \mathbb{E}_{\kappa^*} \left[\int_{\mathbb{R}^p \times \mathbb{R}^p} W_2^2(\rho(y_{k+1}|x_{k+1}), \sigma(w_{k+1}|z_{k+1})) \right] \right\} \quad (35)$$

where $\Delta := (A(s), b(s), B(s), H_{k+1}, d_{k+1}, \tilde{R}_{k+1})$.

For the discrete-time observation model (8), the conditional distribution $\rho(y_{k+1}|x_{k+1})$ is Gaussian and can be written as

$$\rho(y_{k+1}|x_{k+1}) = \mathcal{N}(h(x_{k+1}), R_{k+1}), \quad R_{k+1} \succ 0, \quad (36)$$

whereas for the linear surrogate model (11), the conditional distribution is given by

$$\sigma(w_{k+1}|z_{k+1}) = \mathcal{N}(H_{k+1}z_{k+1} + d_{k+1}, \tilde{R}_{k+1}), \quad \tilde{R}_{k+1} \succ 0. \quad (37)$$

Now, to derive the surrogate linear observation model in (30), we consider the following optimization problem:

$$\min_{H_{k+1}, d_{k+1}, \tilde{R}_{k+1}} \mathbb{E}_{\kappa^*} \left[W_2^2(\rho(y_{k+1}|x_{k+1}), \sigma(w_{k+1}|z_{k+1})) \right]. \quad (38)$$

Proposition 4. *The optimization problem (38) admits a unique solution given by*

$$\begin{aligned} H_{k+1}^* &= \mathbb{E}_{\psi} [(h(X_{k+1}) - \mu_h)(Z - \mu_Z)^\top] \Sigma_{ZZ}^{-1}, \\ d_{k+1}^* &= \mathbb{E}_{\psi} [h(X_{k+1}) - H_{k+1} \mathcal{T}_{X_k}(X_{k+1})], \\ \tilde{R}_{k+1}^* &= R_{k+1}, \end{aligned}$$

where $\Sigma_{zz} := \mathbb{E}_{\psi} [(Z - \mu_Z)(Z - \mu_Z)^\top]$, $\mu_h = \mathbb{E}_{\psi} [h(X_{k+1})]$ and $\mu_Z = \mathbb{E}_{\psi} [Z]$ with $Z = \mathcal{T}_{x_k}(X_{k+1})$. The expectation is taken w.r.t. $d\psi(x_{k+1}, x_k) = d\sigma(x_k) d\rho(x_{k+1}|x_k)$.

Proof. By substituting (36) and (37) into (3), we obtain

$$W_2^2(\rho(y_{k+1}|x_{k+1}), \sigma(w_{k+1}|z_{k+1})) = \|h(x_{k+1}) - H_{k+1} z_{k+1} - d_{k+1}\|^2 + \text{Tr} \left[\hat{\Sigma}_p + \hat{\Sigma}_q - 2(\hat{\Sigma}_p^{1/2} \hat{\Sigma}_q \hat{\Sigma}_p^{1/2})^{1/2} \right], \quad (39)$$

where

$$\hat{\Sigma}_p := R_{k+1}, \quad \hat{\Sigma}_q := \tilde{R}_{k+1}.$$

Substituting (39) in (38) we obtain the following optimization problem in observation model parameters:

$$\min_{H_{k+1}, d_{k+1}, \tilde{R}_{k+1}} \mathbb{E}_{\kappa^*} \left[\|h(x_{k+1}) - H_{k+1} z_{k+1} - d_{k+1}\|^2 + \text{Tr} \left[\hat{\Sigma}_p + \hat{\Sigma}_q - 2(\hat{\Sigma}_p^{1/2} \hat{\Sigma}_q \hat{\Sigma}_p^{1/2})^{1/2} \right] \right]. \quad (40)$$

We then substitute the optimal coupling κ^* from Lemma 3 into (40) and focus on the first term of the objective function, as it is the only part that contains H_{k+1} and d_{k+1} . This yields the following objective function:

$$\tilde{J}(H, d) := \iint d\sigma(x_k) d\rho(x_{k+1}|x_k) \times \|h(x_{k+1}) - H_{k+1} \mathcal{T}_{x_k}(x_{k+1}) - d_{k+1}\|^2. \quad (41)$$

(1) *Optimization w.r.t. d:* Equation (41) can be equivalently rewritten as

$$\tilde{J}(H, d) = \mathbb{E}_\psi \left[\|h(X_{k+1}) - H_{k+1} \mathcal{T}_{X_k}(X_{k+1}) - d_{k+1}\|^2 \right], \quad (42)$$

where $d\psi(x_{k+1}, x_k) = d\sigma(x_k) d\rho(x_{k+1}|x_k)$. We note that $\tilde{J}(H, d)$ is strictly convex. Differentiating (42) w.r.t. d yields

$$\nabla_d \tilde{J}(H, d) = -2 \mathbb{E}_\psi [h(X_{k+1}) - H_{k+1} \mathcal{T}_{X_k}(X_{k+1}) - d_{k+1}]. \quad (43)$$

Setting $\nabla_d \tilde{J}(H, d) = 0$ gives

$$d_{k+1}^* = \mathbb{E}_\psi [h(X_{k+1}) - H_{k+1} \mathcal{T}_{X_k}(X_{k+1})]. \quad (44)$$

(2) *Optimization w.r.t. H:* Let $Z := \mathcal{T}_{X_k}(X_{k+1})$ and define

$$\mu_h := \mathbb{E}_\psi [h(X_{k+1})], \quad \mu_Z := \mathbb{E}_\psi [\mathcal{T}_{X_k}(X_{k+1})]. \quad (45)$$

Substitute $d_{k+1}^* = \mu_h - H \mu_Z$ into (41) yields the following reduced objective

$$\tilde{J}_H(H) := \mathbb{E}_\psi \left[\|h(X_{k+1}) - \mu_h - H_{k+1} (Z - \mu_Z)\|^2 \right]. \quad (46)$$

Taking the derivative w.r.t. H , we obtain

$$\nabla_H \tilde{J}_H(H) = -2 \mathbb{E}_\psi [(h(X_{k+1}) - \mu_h - H_{k+1} (Z - \mu_Z)) (Z - \mu_Z)^\top] = -2 (\Sigma_{hZ} - H_{k+1} \Sigma_{ZZ}). \quad (47)$$

where

$$\Sigma_{hZ} := \mathbb{E}_\psi [(h(X_{k+1}) - \mu_h)(Z - \mu_Z)^\top], \quad (48)$$

$$\Sigma_{ZZ} := \mathbb{E}_\psi [(Z - \mu_Z)(Z - \mu_Z)^\top]. \quad (49)$$

Setting $\nabla_H \tilde{J}_H(H) = 0$ yields the normal equations

$$H_{k+1}^* \Sigma_{ZZ} = \Sigma_{hZ}, \quad \Rightarrow \quad H_{k+1}^* = \Sigma_{hZ} \Sigma_{ZZ}^{-1}. \quad (50)$$

We interpret Σ_{ZZ}^{-1} as the Moore–Penrose pseudoinverse if Σ_{ZZ} is singular. Finally, the second term on the RHS of (39) is the only term that depends on $\tilde{R}_{k+1}$. Using the same arguments as in Theorem 1, we obtain $\tilde{R}_{k+1}^* = R_{k+1}$. $\square$

B. Surrogate Kalman Filter

Using the results from the previous sections, we have obtained a linear system which is a projected version of the nonlinear system using the W_2 metric. This yields the following Kalman-type filter.

a) *Propagation step*: Given the prior $(\mu_{s|s}, \Sigma_{s|s})$, the *prediction* step is

$$\begin{aligned}\mu_{s+\tau|s} &= A^*(s) \mu_{s|s} + b^*(s), \\ \Sigma_{s+\tau|s} &= A^*(s) \Sigma_{s|s} + \Sigma_{s|s} A^*(s)^\top + A^*(s) \Sigma_{s|s} A^*(s)^\top + B^*(s) B^*(s)^\top.\end{aligned}$$

b) *Update step*: Upon observing $y_{s+\tau}$, the *update* step takes the Kalman form with the projected measurement:
Innovation covariance:

$$S_{s+\tau} = H_{s+\tau}^* \Sigma_{s+\tau|s} H_{s+\tau}^{*\top} + \tilde{R}_{s+\tau}^*.$$

W_2 -projected gain:

$$K_{s+\tau} = \Sigma_{s+\tau|s} H_{s+\tau}^{*\top} S_{s+\tau}^{-1}.$$

Update:

$$\begin{aligned}\mu_{s+\tau|s+\tau} &= \mu_{s+\tau|s} + K_{s+\tau} (w_{s+\tau} - H_{s+\tau}^* \mu_{s+\tau|s} - d_{s+\tau}^*), \\ \Sigma_{s+\tau|s+\tau} &= (I_n - K_{s+\tau} H_{s+\tau}^*) \Sigma_{s+\tau|s} (I_n - K_{s+\tau} H_{s+\tau}^*)^\top \\ &\quad + K_{s+\tau} \tilde{R}_{s+\tau}^* K_{s+\tau}^\top.\end{aligned}$$

V. CONCLUSIONS

We have proved two results about computing Gaussian approximations for stochastic dynamical systems. The first one relates to the probability measure obtained from Fokker-Planck equation. The second result corresponds to computing the approximation for the posterior distribution in continuous-discrete filtering problems.

To build on the results presented in this manuscript, an immediate direction is to overcome the suboptimal coupling used in Section IV for computing the posterior. In our current approach, the joint coupling is disintegrated into two parts, and the coupling from the propagation step is reused for the update step. While this is feasible, it is not necessarily optimal. The optimal update-step coupling remains to be computed, either analytically or via computationally tractable methods. Additionally, it would be valuable to compare the resulting filter, obtained by minimizing the distance between the joint state-output distributions, with KL-based variational inference approaches such as those in [9] and [24].

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